

Chapter 15

Argumentation and Proof in the Mathematics Classroom

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This chapter arose out of discussions of the working group on argumentation, logic, and proof and proving in mathematics. It concerns the relationships between argumentation and proof and begins by addressing the question of what we mean by argumentation and whether it includes mathematical proof. For the purposes of education, we regard argumentation as any written or oral discourse conducted according to shared rules, and aiming at a mutually acceptable conclusion about a statement, the content or the truth of which is under debate. It thus includes proof as a special case.

Study of the relationships between argumentation and proof holds great potential for helping teachers and students deal with the tension between the process leading up to the development of a student's proof and the requirements placed on the final

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product. Students need to experience freedom and flexibility during an initial exploratory phase, whilst ultimately producing a proof that conforms to specific cultural constraints involving both logical and communicative norms in the classroom and in the mathematical community.

We also discuss whether the activity of developing proofs under a teacher's guidance can be used to introduce students to meta-mathematical concepts, just as ascertaining the truth of a mathematical statement or the validity of a proof can provide an opportunity for increasing students' mastery of the related mathematical concepts.

Finally, we use examples from the contributions of the members of the working group to illustrate aspects of these issues and suggest some possible activities to gradually increase students' awareness about proving and proof. We also examine the theoretical perspectives and educational issues involved in choosing and designing such activities.

1 Relationships Between Argumentation and Proof

From the time of ancient Greece, philosophers have linked argumentation to three disciplines: rhetoric, dialectic and logic. For the Greeks, rhetoric was the art of effective communication through discourse, especially persuasive public speaking on moral, legal or political issues. Dialectic was the art of conducting a discussion between two or more people who hold differing views but are free to express themselves whilst also seeking to reach agreement. Plato's philosophical dialogues are typical examples of dialectic. Logic was the art of correct thinking, of engaging in well-formed reasoning. Logical argumentation was the verbal manifestation of logical thinking, whether the intended interlocutors were specifically identified or fictitious, or were "ideal" or "universal." Aristotle analysed these concepts in his *Organon*, using *propositions* organised into *syllogisms* as elementary units of discourse, and making a distinction between contingent statements (that could happen to be true, but could also be false) and necessary statements (that could not possibly be false).

1.1 Mathematical Argumentation

Historians (e.g. Boyer 1968) generally agree that the concept of mathematical proof in a form closely related to the modern view was initiated by the Greeks, with Euclid's *Elements* (ca. 300 B. C. E.) as the paramount exemplar. Euclid's treatise starts with a collection of 23 definitions, which are used to state a set of postulates and axioms.¹ Propositions are then derived from the definitions, postulates, and

¹ For a discussion about possible meanings of the terms *axiom* and *postulate* according to the Greeks, see Jahnke (2010).

axioms through a process of deductive reasoning. In Aristotle's terms, the results thus established are "necessary truths." For many centuries, Euclid's *Elements* was used as the principal model of argumentation for the purpose of determining the truth of mathematical statements.

In the seventeenth century, Pascal described the excellence of the Euclidean method for "demonstrating truths already found, and of elucidating them in such a manner that the proof of them shall be irresistible" (Pascal 2007, p. 428). However, his contemporary Descartes complained that "the Ancients" were more careful about proving than explaining (Barbin 1988). Emphasising methods and processes, Descartes wrote that argumentation should be conducted through an analytical process, which would reveal "... the true way along which something has been methodically invented; ... so that a reader eager to follow it ... would understand the point thus demonstrated and make it his own as if himself were the inventor." (Descartes 1961, quoted in Barbin, p. 603; our translation).

In the nineteenth century, with the discovery of non-Euclidean geometries, the role and status of axioms in mathematical theory changed, passing progressively from assertions that cannot be reasonably doubted, to "rules of the game" that provide relations amongst the objects of a theory, which are defined only through the web of relationships thus posited. For mathematicians, research was no longer a matter of discovering a supra-reality that describes an objectively determined world, but rather a question of creating theories that are intrinsically coherent, possibly pairwise inconsistent, each of which would provide a *model* in the sense given that word by Tarski (1944).

The 1950s marked a renewed interest in the formal study of argumentation, culminating with the works of Toulmin (2008) and Perelman (Perelman and Olbrechts-Tyteca 2008). These authors viewed argumentation as a reasoned (and reasonable) discourse about an issue under debate, in which appeals to emotion are excluded but in which rationality need not be conveyed solely through deductive reasoning. Their work was deeply grounded in the study of language, which distinguishes it from epistemology and the methodology of science (Plantin 2005, p. 74). These considerations lead to even further questions about the relations amongst exploration, argumentation, proof, explanation, and justification in mathematics.

Thus, a fundamental tension, probably related to the very nature of mathematical activity, has pervaded the history of mathematical argumentation. Mathematical theories are constructed and developed along multiple paths. Some researchers lay special emphasis on the process of mathematical invention: taking care to keep their minds open to new ways of viewing topics; exploring, through careful analysis, multiple approaches to problems; and gathering insight from alternative perspectives. Other researchers "put energy into making mathematical arguments more explicit and more formal" (Thurston 1994, p. 169): searching for a synthesis to tighten and strengthen mathematical arguments; and for a broad framework into which to fit individual mathematical results. The work of Bourbaki in the twentieth century is certainly an example of the latter approach, as might also be considered the work of logicians such as Frege, Russell, Tarski, and Gödel. For these logicians, it was not only a matter of strengthening the foundations of

mathematics but also of understanding and clarifying the nature of its formalisation at a “meta” level.

Yet a majority of working mathematicians who have written about their own ways of doing research stress the role of formal verification and the logical checking of proofs less than do logicians. For example, in drawing up the list of “brain and mind facilities” researchers use when searching for new results and their justification, Thurston (1994) sets “logic and deduction” side by side with “intuition, association and metaphor,” the importance of which he highlights:

Personally, I put a lot of effort into “listening” to my intuitions and associations, and building them into metaphors and connections. This involves a kind of simultaneous quieting and focusing of my mind. Words, logic, and detailed pictures rattling around can inhibit intuitions and associations. (op. cit., p. 165)

Concerning the phase of checking an argument, Thurston adds, “Reliability does not primarily come from mathematicians formally checking formal arguments; it comes from mathematicians thinking carefully and critically about mathematical ideas” (op. cit., p. 170).

Hadamard and Poincaré (2007) referred to a similar psychological phenomenon. Each described having solutions to problems pop suddenly from their unconscious to their conscious minds, following a period of intense, conscious, but apparently fruitless work. Pólya (1957) also promoted problem-solving methods in which formal deductive reasoning is not the chief consideration. Thom (1974) went further, linking the psychological aspects of the research process to the epistemological role of *meaning* in mathematics, which he referred to as “the ‘ontological justification’ for mathematical objects” (op. cit., p. 49).

However, the significance these mathematicians gave to the intuitive aspects of their research activity may proceed from their desire to rectify the rigid, formalistic image often associated with the field. None of them rejected the importance of careful deductive reasoning. For example, Thurston (1994) wrote,

We have some built-in ways of reasoning and putting things together associated with how we make logical deductions: cause and effect (related to implication), contradiction or negation, etc. (op. cit., pp. 164–165).

The importance of rigorous verification of arguments to mathematicians was made clear in the 1990s following the announcement in June 1993 of a proof for Fermat’s Last Theorem, which was reported on the front pages of newspapers around the world. Although Andrew Wiles had worked through many of the details of his “proof” with his colleague Nicholas Katz during the months before the dramatic announcement, it was only during the summer and fall of 1993, when he was writing out the proof for publication, that he found a mistake. After almost a year of painstaking effort, during which he and an associate carefully examined every step of the argument, Wiles finally realised that an approach he had previously thought to be a dead end was, in fact, the way to complete the proof (Singh 1997).

Thus, to varying degrees depending on the mathematician and the particular circumstances of the work, it appears that mathematical activity integrates two approaches. Alcock (2009) describes one as *semantical*: the study of examples

and counterexamples; experimentation; instantiation of concepts, properties, and theorems; and development of analogies, associations, mental images, metaphors, and so forth. She describes the other approach as *syntactical*: use of formal definitions and symbolism; manipulation of formulas; and production of inferences that have been checked for logical correctness. Wilkerson-Jerde and Wilensky also commented upon this duality of approaches as they observed mathematicians attempting to solve problems they had not seen before:

For some experts, specific instantiations of the mathematical object being explored serve a central role in building a densely-connected description of the proof; while for others a formal definition or several small components of the mathematical object serve this purpose. (2009, p. 271)

Barrier et al. (2009), Blossier et al. (2009) and Durand-Guerrier and Arsac (2009) proposed that the mathematical work done when searching and developing proofs is essentially dialectical, with frequent movement back and forth between semantic exploration of the mathematical objects under consideration and more formal syntactical analysis of their definitions, properties, theorems, and formulas. To support their thesis, these authors cite examples from Bolzano, Liouville, Cauchy, Euclid, and the Pythagoreans.

The dividing line between semantics and syntax is seldom clearly fixed for individuals; it evolves through their activities and also depends on their levels of mathematical maturity and mastery of formalisms. Fischbein (1982) suggests that the related *intuitions* also evolve, being a springboard for argumentation and proof and simultaneously being enriched, refined and strengthened by them. Moreover, individuals might themselves regard their activities as semantic, whereas an observer might regard them as syntactic, and vice-versa.

1.2 *Argumentation, Proof, and Proving in Mathematics Education*

The tension between the syntactic and semantic aspects of mathematical argumentation for mathematicians has its counterpart for mathematics educators. Didacticians (e.g., Balacheff 1987; Duval 1991; Hanna 1989) have pondered the relationships and potential oppositions involved: between proof as a cultural product subject to constraints of consistency and communication and proving as the process aiming at that product; between mathematical outputs that must fit some rules (i.e., some logical and textual models) and the creative and constructive side of mathematicians' activity; and between mathematical proof and ordinary argumentation. They have considered, *inter alia*, whether rigorous proof and ordinary argumentation create obstacles for each other and which educational conditions would foster fruitful use of either or both of the two approaches.

For example, Duval (1991) argued that (informal) argumentation impedes the learning of (mathematical) proof. He claimed that in argumentation propositions

assumed as conclusions of preceding inferences or as shared knowledge are continuously reinterpreted (ibid., p. 241), with primary importance being placed on their semantic content. By contrast, in a deductive step in a mathematical proof, “propositions do not enter directly by reason of their content, but rather through their operational status” (ibid., p. 235; our translation). In other words, because it is a deductive chain of inferences, similar to a chain of calculations, the structure of a proof strongly depends on the operational status of the propositions that compose it. Now, in a given inference (or deductive step), the operational status of a proposition – whether premise, *énoncé-tiers*² or inferred proposition – is independent of its content, since a proposition may change its status within the same proof: for instance, when an inferred proposition is “recycled” as a premise for the next deductive step. The possible operational status of the propositions involved in an argument are differentiated according to their *theoretical epistemic value*; namely, as hypotheses or previously inferred propositions for the premises, and as definitions, theorems, or axioms for the *énoncé-tiers*. Duval argues that students have difficulty in learning mathematical proof because they do not easily grasp its specific requirements and hence deal with it as simple argumentation.

Other authors (e.g., Bartolini Bussi 1996, 2009; Grenier and Payan 1998; Lakatos 1976) studied the activity of proving from a *problem-solving* perspective. Some, such as Bartolini Bussi (1996, 2009), Bartolini Bussi et al. (1999) and Douek (2009), have differed with Duval (1991), claiming that argumentation can be used as an effective basis for mathematical classroom discussions concerning not only mathematical reasoning in general but also the rules for mathematical proof and the mathematical practices that play important roles in proof production. These include changing the frame of reference (e.g., from geometric to algebraic), using different representation registers, introducing a new object (e.g., a figure), and reviewing the role played by definitions in the production of a proof (cf. Durand-Guerrier and Arsac 2009). Written proofs, of course, do not include discussion of the practices that led up to their production.

The problem-solving perspective considers a “written proof” as the outcome of a process in which it is helpful to take fully into account the initial and intermediate phases. These phases vary, depending on how the problem is presented to the students; in particular, whether the result is stated or must be discovered; and, if stated, whether its truth-value is given (“Show that...”) or is left to the students to determine. Students asked to find a result or to establish its truth-value start the process with an exploratory phase, in which they use various heuristics, such as studying examples, searching for counterexamples, and reflecting on definitions and theorems that may be linked to the problem. This phase normally leads to the statement of a conjecture.

² An *énoncé-tiers* is a statement already known to be true – namely an axiom, a theorem, or a definition – and generally is (or could be put) in the form “if p , then q ”. In Toulmin’s (2008) model (see below), an *énoncé-tiers* corresponds to a warrant in an inference step.

Barrier et al. (2009) described and analysed the relation between the exploratory phase and the organisation of a proof by extending the concepts of *indoor* and *outdoor* games, introduced by Hintikka (1996). They defined outdoor games as concerned with determining the truth of statements within a given interpretative domain, in which the mathematical objects, their properties, and their relationships are investigated and tested. By contrast, indoor games are concerned with establishing the validity of statements inside a given theory; they consist of strategically using mathematical properties and assertions, such as hypotheses, axioms, and theorems available in the theory, in order to combine them syntactically into a proof, using computation, symbolical manipulations, and deductions. Barrier et al. argued that the construction of a proof most often rests upon a back-and-forth process between outdoor games and indoor games, and that semantic (outdoor) and syntactic (indoor) reasoning are both necessary and dialectically related. In addition, Blossier et al. (2009) proposed that quantification, whether explicit or implicit, is crucial for the semantic control realised through outdoor games. Epp (2009) also noted its importance, pointing out and describing students' difficulties with quantification in various contexts involving proof and disproof. Finally, Inglis et al. (2007) described an experiment where an advanced student eventually succeeded in developing a proof through a dialectical interplay between examination of numerical examples and syntactical analysis of definitions.

To characterise the possibility that heuristic argumentation can result in a conjecture with arguments or 'reasons' suitable for constructing a proof, Garuti et al. (1996) introduced the notion of *cognitive unity*, giving evidence for it in an experiment involving 36 students (two classes) in grade 8, where the arguments students gave in support of their conjecture about the "geometry of sun shadows" are essentially the same as those they arranged into a proof. (See also Boero et al. 2007). Pedemonte (2007) showed experimentally (with two geometry theorems proposed to 102 students in grades 12 and 13) that in some cases where cognitive unity seems possible, some students encounter difficulties in constructing a deductive chain from the otherwise suitable arguments that they produced through exploring and conjecturing, especially when the conjecture is obtained through abduction. She then distinguished between *cognitive unity* (concerning the referential system) and *structural continuity*. Structural continuity between argumentation and proof occurs when inferences in argumentation and proof are connected through the same structure. Pedemonte (2008) also used the notion in examining the same students' algebraic work, showing how in that case cognitive unity more easily results in structural continuity. As concerns cognitive unity and structural continuity, the educational and didactical context in which students are asked to produce conjectures and justifications seems crucial. In particular, tasks designed to lead to a conjecture, which include a request for providing reasons to support it, enhance argumentative activities that may facilitate a link between conjecturing and proving.

In order to study more precisely the articulation between argumentation and proof, Douek (2009) made use of Lolli's analysis of proof production (cf. Arzarello 2007). She considered three modes of reasoning: heuristic exploration, reasoned organisation of relevant propositions, and production of a deductive text obeying mathematicians'

norms. Her detailed analysis of these modes showed that producing a proof depends not only on deductive reasoning but also on other activities, almost all dependent primarily on semantic content, including exploration using inductive reasoning, empirical verification of results, and abductive reasoning and argumentation. One difficulty of teaching proof is that the different modes of reasoning obey different rules of validity. For example, exploration that relies solely on examples cannot substitute for the kind of argument needed to establish the truth of a general mathematical proposition. According to Douek, effective use of the different modes of reasoning requires a certain amount of meta-mathematical reflection about the “rules of the game” of proving.

In an endeavour to find a unifying frame to analyse all these aspects of argumentation and proof, authors such as Pedemonte (2007), Inglis et al. (2007), Arzarello et al. (2009a), and Boero et al. (2010)³ refer to the work of Toulmin (2008). Toulmin proposed a ternary model for the inference step in argumentation, which consists of a Claim, Data, and a Warrant, along with auxiliaries such as a Qualifier, a Rebuttal, and Backing. Use of this model allows one to compare argumentation and mathematical proof. For proof, the warrants (and ultimately the “backings”) come from a mathematical theory,⁴ whereas for argumentation they may consist of visual evidence or properties established empirically, for example.

Considering Toulmin’s (2008) model and Duval’s (1991) thesis together raises a fundamental didactical problem: namely, how to make students aware that when the final product is to be a “proof,” the warrants must be chosen from amongst propositions with a precise theoretical status and selected according to specific criteria. Balacheff (1987) suggested a necessary shift from the pragmatic to the theoretical. Tanguay (2007) described the necessity to refocus students’ outlook from *truth* to *validity*: students must understand that it no longer matters whether the involved propositions are intrinsically true or false but whether a sequence of deductive steps is valid. Barrier et al. (2009) and Durand-Guerrier (2008) go still further, considering how students need to coordinate their ideas of truth in an interpretative domain and the concept of validity in a theory. The didactical and pedagogical questions concerning how to move students towards such a shift are addressed in the next section.

2 Paths to Constructing Argumentation and Proof in the Mathematics Classroom

The emergence of proof reflects a variety of cultural characteristics. Amongst these is a desire for justification, for providing rational support to back up arguments, and for organising processes algorithmically. Attempting to assess the extent to

³ More precisely, the frame proposed by Boero et al. (2010) results from the combination of Toulmin’s model with Habermas’ (2003) criteria of rationality.

⁴ This agrees with Mariotti’s definition of “theorem” (Mariotti et al. 1997): a system consisting of a statement, a proof – derived from axioms and other theorems according to shared inference rules – and a reference theory.

which argumentation in a wide range of disciplines constitutes logically valid proof is an important means of examining the grounds on which arguments and inferences are conducted in the “real world.” It also promotes understanding for the nature and specific qualities of mathematical proof and their connections to various mathematical fields.

Therefore, not only mathematical theories but also the various modes of reasoning used in mathematics are important for students’ intellectual development. Especially at the beginning, students are not exposed to any meta-mathematical reflection concerning the terms used in mathematics or to the differences from their ordinary uses and the systemic connections amongst them. As Vygotsky (1986) put it, these ideas are not treated as “scientific concepts” (i.e., as an explicit area of scientific knowledge), whereas organised exposure to them could lead students to master argumentation that extends beyond mathematics to science or further (e.g., to law; cf. Perelman and Olbrechts-Tyteca 2008). Students could thus benefit from familiarity with the wide variety of mathematical, ‘meta-’ and ‘extra-’ mathematical practices linked to the activity of proving.

2.1 What Do We Want Students to Experience?

Exploration, validation, and interpretation engender students’ need for understanding when they are confronted with unexpected results, contradictions, or ambiguities. Complex tasks, which lead to doubt because of possible contradictions or obvious ambiguities (through solo activities or with peers), produce a didactic situation that favours proof and stimulates the development of several aspects of its practice. By contrast, the simplified tasks all too frequently proposed to students may obscure the need for proof and proving. Similarly, asking for proofs of statements that appear obvious to students’ perception (often the case in middle-school geometry) or of which neither the meaning nor the truth is questioned may have a detrimentally effect on students’ understanding of the nature and purpose of proof.

For example, students are rarely, if ever, presented with false mathematical statements and asked to determine whether or not they are true, or under what conditions, they might become so. In fact, it is desirable for students to experience uncertainty about (mathematical) objects, statements, and justifications in order to cultivate an attitude of reasonable scepticism. Problems whose truth is in question or open-ended problem situations require students to engage in exploration and hence argumentation (e.g., Arsac and Mante 2007; Grenier and Payan 1998; Legrand 2001).

An emphasis on exploration naturally generates a need to reflect on the place of conjectures. Whether for mathematical reasons (e.g., lack of cognitive unity between the thought processes leading to the conjecture and those required for a proof) or for curricular reasons (cf. Tanguay and Grenier 2009, 2010), conjecturing does not necessarily prompt proving. Crucially, conjecturing should go beyond mere guessing or unbridled speculation and encompass in some way a search for a structural explanation. Only then does it involve meaningful exploration and interpretation and a genuine need for validation.

Previous experiences of doubt about the truth or falsity of mathematical statements can lead students to see the need for validation as meaningful in terms of their own experiences, conjectures, and mathematical backgrounds, as well as in ways that they have shared with others in their class. Also important is the need for students to realise that various levels of validation are appropriate in various situations, which depend on context, the norms of practice of the community, their own and others' reference knowledge, and so forth. In addition, some important meta-mathematical concepts and their systemic interconnections, especially the relationships between axioms and validity of arguments, can be developed through activities using statements that are valid in one "theory" and not in another, and yet are accessible to a broad range of students (e.g., Parenti et al. 2007). Argumentation within class deliberations about what is acceptable (or not) and under what circumstances may open the way towards a more theoretical way of thinking; appreciation for theory may be enhanced by meta-mathematical discussions. These class discussions can address such questions as "Why do we prove?" and "What does it mean to have proved something?" as well as the relation to axioms, definitions, and more broadly the "rules of the game." For example, what is meant by "checking a proof" or "checking a program" may concern the mode of reasoning (whether deduction, induction, or abduction), the respective roles of syntax and semantics, the relationship between truth and validity, and the logic relevant to the given context.

In addition, in learning about theorems and their proofs, students need to experience not only how to validate statements according to specific reference knowledge and inference rules within a given theory, but also how the "truth" of statements depends on definitions and postulates of a reference theory (e.g., in geometry; cf. Henderson 1995).

Later, when reflection on meta-mathematical concepts needs to become deeper, more exhaustive, and more systematic (e.g., in higher secondary school or in the education of mathematics teachers), tasks could be employed that illustrate the central role definitions play in the formulation of statements, in argumentation, and in proof.

Meta-mathematical issues such as the variability of the means and tools for validation, the conventions for definitions, and the ambiguity of everyday speech compared with the precision of mathematical language, underline the central role of interpretation in argumentation and proof. Interpretation calls for various mathematical skills, such as the ability to change the semiotic register or frame of reference; relate certain mathematical objects to others; use theories, whether mathematical or not; use analogies and metaphors; and transfer procedures from one problem situation to another. At a higher, more specialised, level, we can explain these issues in terms of Tarski's (1983) theory of logical interpretation as the process of validating a mathematical construction by embedding it into a *mathematical model*. More pragmatically, they represent instances of going back and forth from a general statement to examples (whether generic or particular), relating schemas to situations, and referring to a precise context to understand a general statement. Students should experience the need to interpret figures, expressions, statements, and representations in various semiotic registers and develop virtuosity in doing so. Such activity

would help make them aware that one cannot take effective communication for granted in mathematics.

Developing competence in *communication* is important, but exercising it relies on the presence of one or more listeners to react, contradict, and make use of what is communicated. Success in communicating in the classroom also depends on consciously shared references and on some form of shared theoretical scaffolding, which enhances the importance of debating mathematical issues in the classroom. However, as Duval (2001) pointed out, the form of control needed for (mathematical) proof development requires that students be able to figure out connections or simultaneities between propositions remote in the discourse, to come back to propositions already stated, to reconsider how propositions are organised, and to allow pauses and reflection. All of these activities only take place to the fullest extent when a *written* product is required.

In sum, students need to experience two main practices: a divergent exploratory one and a convergent validating one. They would then become familiar with the openness of exploration, that is, its “opportunistic” character and its flexible validation rules. They would also learn the rigorous rules needed to write a deductive text and the strict usage of words, symbols, and formulas when constructing or organising a theory. Previous curricula have enhanced the latter, but more recent ones have swung towards the former. However, isolating one to the detriment of the other weakens both, because they are related dialectically. Besides, once we consider exposing them to a wide range of experiences, it is crucial to help students face the contradictory differences amongst the diversity of practices used in argumentation and proof and to make them aware of when and how they can use these practices.

2.2 *Designing Learning Environments and Activities*

Here, we discuss various classroom activities aimed at developing both mathematical and meta-mathematical concepts. The majority were presented at the 19th *ICMI Study Conference, Proof and Proving in Mathematics Education*.

In one example, Arzarello et al. (2009a, b) described a long-term teaching experiment extending over the 5 years of Italian secondary school (grades 9–13). The experimenters gradually introduced students to local linearity through graphical, symbolic and numerical exploration using extensive technology, group work and class discussion. The problem-solving activities were intended to foster cognitive continuity between the exploration and production phases of developing a proof. A main result of the particular experiment reported in Arzarello et al. (2009a) was the creation of explicit links between the pieces of knowledge that were involved.

Perry et al. (2009a, b) implemented an innovative course for pre-service high school mathematics teachers, which features the collaborative construction of an axiomatic system involving points, lines, planes, angles, and properties of triangles and quadrilaterals.

Tanguay and Grenier (2009, 2010; cf. Dias and Durand-Guerrier 2005) reported an experiment conducted with preservice teachers in France and in Quebec, to whom the following classroom situation was submitted: (a) define and describe regular polyhedra; (b) produce them with given materials; and (c) prove that the list produced in (b) is complete. Very few of the students could conceptualise the relationship between the angles of the faces and the dihedral angles in a polyhedron in a way that would have accessed a proof that only five regular polyhedra exist. Instead, the students uncontrollably used intuition and analogy; some erroneously believed in an infinite sequence of regular polyhedra (therefore, e.g., trying to construct a regular polyhedron with hexagons). This error may have resulted because in a typical classroom proof and verification are closely associated with algebra and manipulation of formulas, with conjectures viewed empirically rather than as necessitating structural explanations.

Boero et al. (2009) suggest that random phenomena can provide teachers with opportunities to deal with definitions as well as more subtle aspects of mathematical argumentation and proof. For example, in spite of its potential logical and theoretical weaknesses, the classical definition of probability can be used in primary school as a basis for approaching the notion of an event's probability and of quantitatively estimating it on the basis of the search for an appropriate set of equally-likely cases. In high school and at the university level, the definition can be used for reflecting on the requirements of a good definition within a theory and related meta-mathematical issues concerning proof.

Brousseau (1997) described a didactical situation in which students aged 9–10 were given the task of enlarging the size of a jigsaw puzzle. Engaging in the task led them to discover for themselves that adding the same amount to the dimensions of all the pieces did not work because the pieces no longer fit together! Of course, at primary school, students cannot provide a mathematical proof that multiplication rather than addition is necessary; the teacher is responsible for asserting the result. Nevertheless, the situation promoted students' understanding that the result did not come from an arbitrary decision by an authority and helped them to take responsibility for their own learning. The activity was originally developed in the 1970s by Brousseau in France as part of a study on the teaching of rational numbers and finite decimals at primary school, and the results were subsequently reproduced many times.

In particular, secondary and tertiary level students have significant difficulty in developing an understanding about the nature of the set of real numbers (cf. Barrier et al. 2009). In an earlier experiment, Pontille et al. (1996) analysed the work of several groups of 3–4 students in grade 11. One group was followed for the entire year. The students were asked whether every increasing function from $\{1, 2, \dots, n\}$ to $\{1, 2, \dots, n\}$ has a fixed point. Once they had resolved this problem, they were asked to study generalisations to functions defined on the set of all finite decimals (i.e., real numbers with finite decimal expansions) in the interval $[0;1]$, and then on or any subset of the real numbers. The students conducted their work outside of the classroom, under the supervision of a researcher. They wrote results, questions, conjectures and proofs in notebooks, which the researchers analysed along with student interviews.

Students initially managed to solve the problem for the integers, using proofs that relied strongly on the fact that every natural number has a successor. When they then considered the further problems, they encountered significant epistemological questions: (a) Is there a fixed distance between any two “consecutive” finite decimals (i.e., can the notion of successor be used here?) (b) Given any positive finite decimal, can one find a smaller such number?⁵ (c) Is there a relation between questions (a) and (b)? The students finally concluded that there is no such thing as a smallest positive finite decimal or a fixed distance between any such number and its ‘successor.’

The students then examined a more complex problem involving the graph of an increasing function. Through discussion, the students developed various representations of the numbers involved (e.g., representing two kinds of numbers with different-coloured squares; drawing a straight line graph that contained holes). After a number of exchanges, they found a solution, which also led them to discover a counterexample.⁶ Pontille et al.’s description of the students’ work shows that students can develop reasoning ability and mathematical knowledge in tandem. It also shows that they may need a long period of intensive work before finding a counterexample. That discovery only occurred following the students’ reconsidering, reorganising and adapting their knowledge about real numbers. The research supports the desirability of giving students tasks that lead them back and forth between formal and informal approaches, thus deepening their understanding of mathematical objects whilst simultaneously challenging their reasoning powers. Taking into account both the syntactic and semantic aspects of mathematical proof, such tasks promote the development of both mathematical concepts and meta-mathematical concepts.

In Italy, the Modena group led by M. Bartolini Bussi and the Genoa group led by P. Boero have performed several long-term teaching experiments, in Grades 4–8, to introduce students to validation and proof on both the practical and theoretical levels.⁷ In order to implement their experiments, they attempted to construct suitable contexts, easily accessible or already familiar to students, where postulates represented basic “obvious” properties but where statements to be validated were not obvious and required reasoning.

For example, Parenti et al. (2007) reported a teaching experiment, performed with 80 students by three teachers in four classes from Grades 6–8, which used the potential of two contexts: the context of the *representation of visual spatial situations* in Grades 6–7 (cf. Bartolini Bussi 1996) and the context of *sun shadows* in Grade 8 (cf. Garuti et al. 1996). In both cases, students moved from a familiar

⁵ Notice that if we consider decimal numbers with a given fixed number of digits following the decimal point, then these two questions are answered in the affirmative.

⁶ The activity was also given to French students at a more advanced mathematical level, where it gave rise to very similar results.

⁷ See Hsu et al. (2009) for another example of an instructional experiment concerning validity.

context (e.g., sun shadows) to the geometry of that context and to some of its obvious properties. (E.g., in the geometry of *sun shadows*, straight lines are projected onto a plane as straight lines or points). In both cases, students learned to infer statements from the properties that seemed obvious to them and to move progressively towards the format of standard proofs. At the end of the experiment, in Grade 8, students came to realise that statements validated in the *sun shadows* geometry may be invalidated in the *representation of visual spatial situations* geometry. They also discovered that proof in *sun shadows* geometry was based on an “obvious property” (parallel straight lines are projected onto a plane as parallel or coincident straight lines or points) which in general is not acceptable in the *representation of visual spatial geometry*.

Thus, the students learned that one can regard spatial situations from more than one point of view, each of which may lead to valid conclusions in its own terms. Regarding systems of postulates as obvious properties of spatial representations according to specific ways of thinking about them not only recalls the discovery of non-Euclidean geometries but also resonates with some recent epistemological positions concerning the close connections between geometric axiomatics and related ways of seeing and thinking about space (cf. Berthoz 1997; Longo 2009).

3 Management, Didactical Organisation, and Teachers’ Roles

The preceding examples of activities focused on the development of mathematical and meta-mathematical concepts rather than on how students were asked to carry out their work. However, when aiming to develop competence in argumentation and proof, one needs to carefully organise student-teacher interactions and activities.

A main challenge in teaching argumentation and proof is to motivate students to examine whether and why statements are true or false. Mathematics educators broadly agree that we need alternatives to traditional tasks, which ask students to provide proofs for statements already presented as true. Thus, many mathematics educators now promote the development, at every level of the curriculum, of problems where the truth-goal is at stake. Such problems fall into two categories: problems used to introduce and develop a mathematical concept and to prove theorems about it; and problems used to develop competence in argumentation and proof.

Recently, a number of mathematical educators have turned their attention to developing theoretical frameworks for developing teaching experiments that involve students in solving such problems (cf. Lin et al. 2009).

Given increasing scepticism about the once common view that mathematics teachers should simply deliver mathematical truths, Brousseau (1997) created a set of situations leading to the development of all the required knowledge about finite decimals in elementary school. He also provided a theoretical framework for his work whereby students develop intended knowledge through collective engagement in a succession of phases involving action, formulation, validation, and institutionalisation.

Similarly, in France in the 1980s, the concept of *scientific debate* was introduced into tertiary-level mathematics courses so that students would come to feel personal responsibility for the conjectures they formulate (Alibert 1988). Instructors led students to consider themselves as a community of mathematicians (a specialised instance of a *community of practice*; cf. Wenger 1998) needing to engage in scientific debate on topics proposed by the teacher. They were invited to formulate conjectures and take sides as to the relevance and the truth of these (Legrand 2001).

In another theoretical framework (the *didactic of fields of experience*), students' written solutions and related discussions play a crucial role (Boero et al. 2007, 2008, 2009).

Tanguay and Grenier (2009) pointed out that "implementation in the classroom or in the curricula of the findings of didactical studies are almost always accompanied by alterations and even distortions." Effective implementation of the insights from such studies as those above poses a complex challenge for the designers of teaching experiments.

To take one example, Douek (2009) implemented many of the principles discussed above in experiments in a class of 27 students aged 13–14 in a French middle school. Her aim was to involve students in reorganising the arguments in a proof through discussion about their various roles. Theorems with no cognitive unity had been chosen, in order to allow the teacher to put some features of proof into evidence whilst guiding the process of proving. Argumentation leading to a conjecture was promoted through classroom debate (cf. Bartolini Bussi 1996), and then the experimenter, after leading the students through a proof, encouraged them to see its rationale as meaningful by making a story out of the steps in their reasoning and calculations, and to express themselves both by explaining and by raising questions. Creating a story was intended to help students connect the various steps with the reasons that supported the conclusion, sort the 'blocks' of reasoning into which the proof was divided, and explain how the blocks were joined logically and their possible hierarchical relationship (Knipping 2008). It also provided students with opportunities for personal expression and internalisation of newly-encountered mathematical practices.

For example, in a class session about the Pythagorean theorem, several tasks led students to conjecture the theorem. Then, they had to prove their conjecture by verifying statements prepared by the teacher. Six students were able to prove all the statements. A follow-up task invited students to review the reasons for the steps they had been led through. It triggered class discussion, guided by the teacher, that identified why the statements were needed, and how they were linked together to form a general argument. Some blocks used abductive reasoning (What property makes such-and-such claim possible? How can you view A so that such-and-such relation appears?). Others used deduction. Therefore, mathematical proof practices, such as the different types of logical reasoning used and movements from geometric to algebraic analysis, became a significant part of the discussion. Then the students were asked to write, individually, the story of the proof. Eleven wrote reasonably coherent stories, but only two explicitly described why all the steps were needed and useful. The experiment showed that some students can accept the logical flow of a proof but lack consciousness about the reasons that enable it.

4 Conclusion

In the first part of this chapter, we tried to elucidate the complex relationships between argumentation and proof in mathematics from mathematical and educational perspectives, noting the general lack of consensus about the interrelations between the two, both amongst mathematicians and amongst mathematics educators. In the second part, we examined the aims, conditions, and constraints that affect planning and implementation of appropriate situations for facilitating argumentation and proof in mathematical classes. First, we considered what we want students to experience and what aspects of learning proof should be the goal of instruction. We need to develop students' awareness about the core aspects of theorems and proofs and thus support their use when needed. We must engage students in mathematical tasks that also foster the development of meta-mathematical concepts. In the third part, we presented short descriptions of examples of learning environments supportive of the goals for the crucial activities of proving identified in the second part. In sum, engaging students in situations which make them aware of the constructive character of mathematical activities, especially those involving conjecture and proof, poses complex challenges. One needs to consider:

1. The types of tasks: open-ended problems, such as statements whose truth or falsity students are to determine; critical study of students' arguments; discussion about the basis for justification; story-telling; work on proof development; and so forth;
2. Management of activities: for example, student discussions; production of written work; use of software, diagrams, and instruments; class management techniques: use of reasoning diagrams; meta-level discourses; managing different semiotic registers; whole-class debate; working in groups; working individually; and so on;
3. The respective roles of students and teachers.

We hope that we have succeeded in making clear the various perspectives underlying the educational proposals, giving insight into their richness, and providing elements for further research and innovation.

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